

Garage Management System – Salesforce CRM

College Name: Terf's Academy College of Arts and Science

College Code: BRU5E

Team ID: NM2025TMID27505

Team Members

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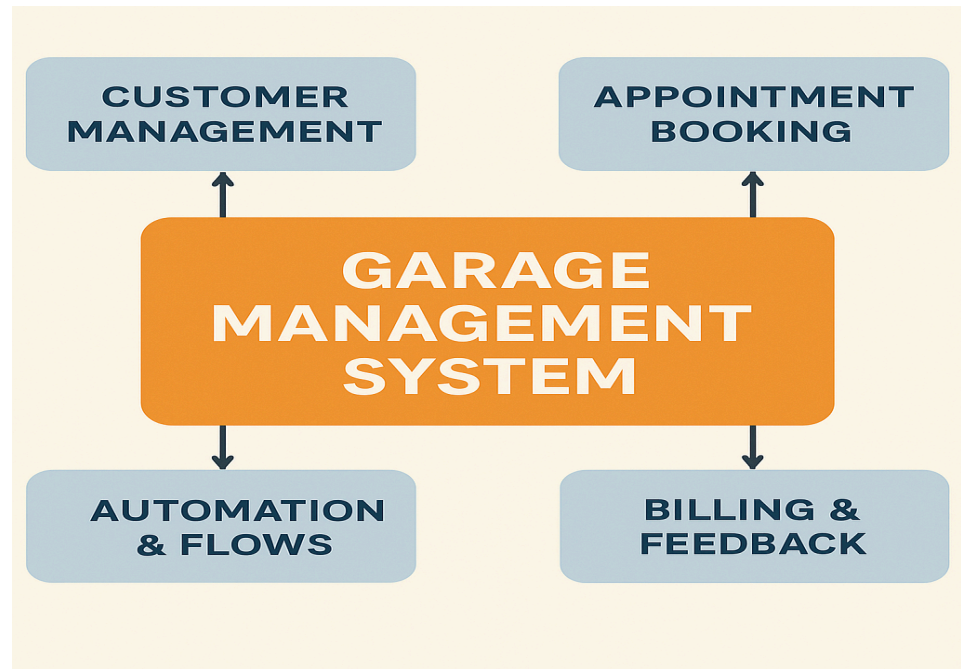
1. Project Overview

The **Garage Management System (GMS)** is a **Salesforce-based CRM** designed to simplify and automate automotive service operations.

It helps garages efficiently manage:

- Customer details
- Appointment scheduling
- Service records
- Billing and feedback collection

This system improves **operational efficiency**, ensures **quality service delivery**, and **enhances customer relationships** by leveraging Salesforce features like automation, dashboards, and reporting.



2. Objectives

The main objective of this CRM is to **automate and streamline garage operations**. By integrating Salesforce CRM features, the system enables:

- Better **customer management** and engagement
- Improved **service tracking** and real-time monitoring
- Faster and **error-free billing & payment** processes
- Enhanced decision-making with **visual dashboards and reports**

Business Value:

This system **reduces manual errors**, **improves productivity**, and **increases customer satisfaction**.

3. Development Phase

Creating Developer Account

A **Salesforce Developer Account** was created using this URL:

<https://developer.salesforce.com/signup>

Build enterprise-quality apps fast and get hands-on with Agentforce and Data Cloud.

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- ✓ Build apps fast with drag-and-drop tools
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- ✓ Ground Agentforce with structured and unstructured data
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Sign up for your Developer Edition

A free Salesforce Platform environment with Agentforce and Data Cloud

First name: Dharani ✓ Last name: S ✓

Job title: Developer ✓ Work email: dhaaranishanmugan ✓

Company: Terf's Academy ✓ Country/Region: India ✓

Your org may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

☒ I agree to the Main Services Agreement – Developer Services and Salesforce Program Agreement. I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

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☒ I'm not a robot

Created Custom Objects

The following **custom objects** were developed to manage garage operations:

- **Customer Details** → Stores customer information.
- **Appointments** → Tracks all customer service bookings.
- **Service Records** → Maintains detailed service history.
- **Billing & Feedback** → Captures payment details and customer ratings.

Object Manager

52+ Items, Sorted by Last Modified

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Billing details and feedback	Billing_details_and_feedback_c	Custom Object		8/28/2025	✓
Service records	Service_records_c	Custom Object		8/28/2025	✓
Appointment	Appointment_c	Custom Object		8/28/2025	✓
Customer Details	Customer_Details_c	Custom Object		8/28/2025	✓

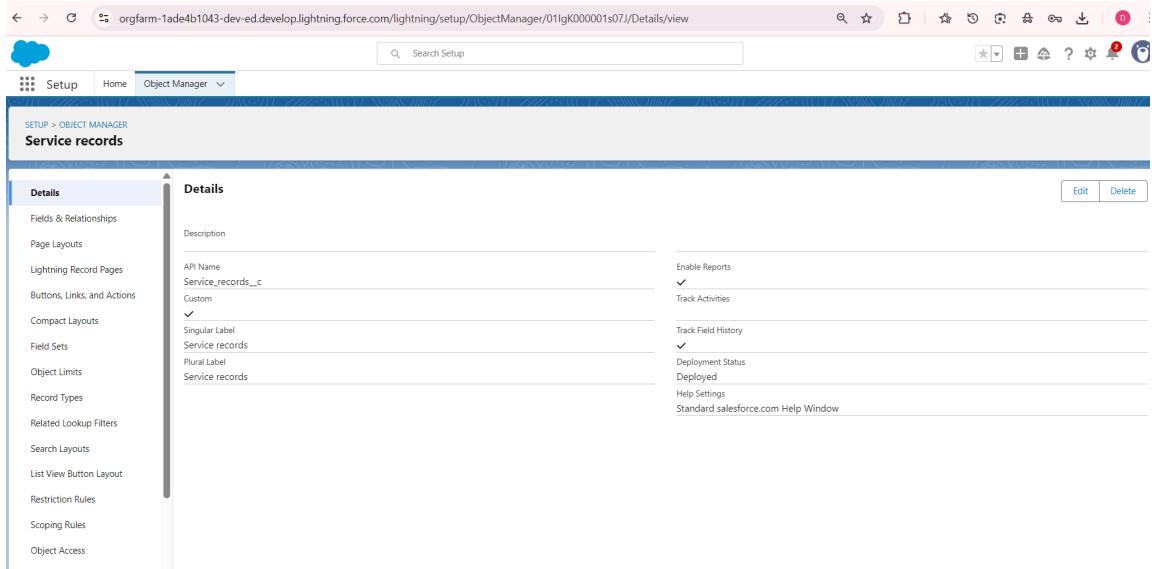
Configured Fields and Relationships

- **Lookup Relationship: Appointment → Customer Details**

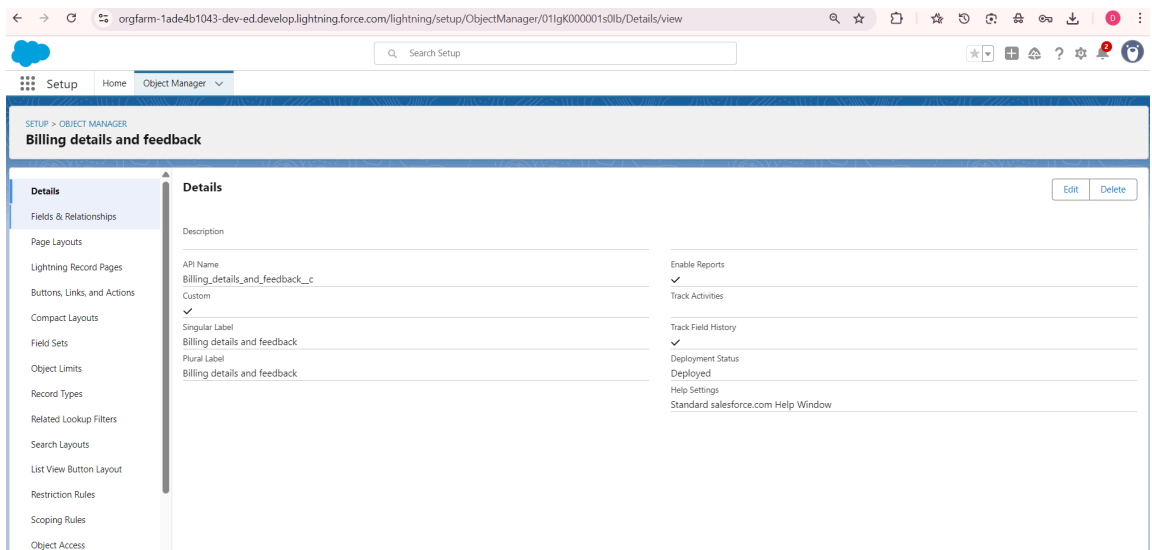
The screenshot shows the Salesforce Setup page for the 'Customer Details' object. The left sidebar contains a list of configuration options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, Object Access, and Triggers. The 'Details' section is selected, showing a form with the following fields: Description, API Name (Customer_Details__c), Custom (checked), Singular Label (Customer Details), Plural Label (Customer Details), Enable Reports (checked), Track Activities (checked), Track Field History (checked), Deployment Status (Deployed), and Help Settings (Standard salesforce.com Help Window). The 'Edit' and 'Delete' buttons are visible in the top right corner.

The screenshot shows the Salesforce Setup page for the 'Appointment' object. The left sidebar contains a list of configuration options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, Object Access, and Triggers. The 'Details' section is selected, showing a form with the following fields: Description, API Name (Appointment__c), Custom (checked), Singular Label (Appointment), Plural Label (Appointments), Enable Reports (checked), Track Activities (checked), Track Field History (checked), Deployment Status (Deployed), and Help Settings (Standard salesforce.com Help Window). The 'Edit' and 'Delete' buttons are visible in the top right corner.

- **Master-Detail Relationship: Service Records → Appointments**



- **Lookup Relationship: Billing & Feedback → Service Records**

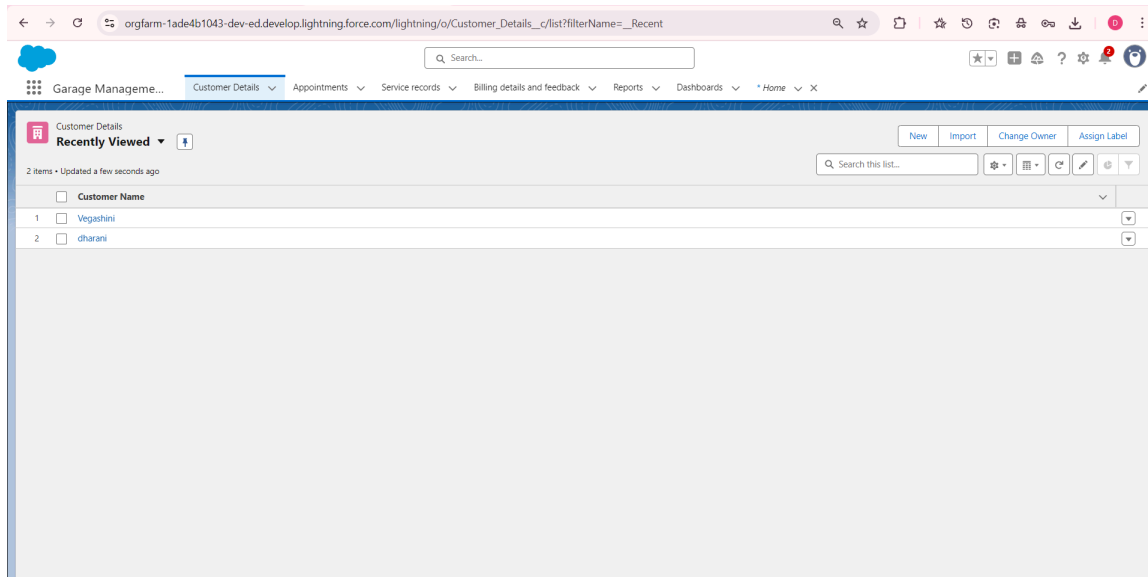


Developed Lightning App

A **Lightning App** named **Garage Management System** was created, which includes the following tabs:

- Customer Details

- Appointments
- Service Records
- Billing & Feedback
- Reports & Dashboards



Implemented Flows

Automation flows were developed to:

- **Auto-update** the billing amount after payment completion.
- Automatically change **service status** to “**Completed**” after a quality check.

Validation Rules

- **Appointments:** Validates **vehicle number plate format**.

Garage Manageme... Customer Details Appointments Service records Billing details and feedback Reports Dashboards * Home X

Appointment app-009 New Contact Edit New Opportunity

Related Details * = Required Information

Appointment Name app-009 Owner Dharani S

Customer Details Vegashini X

* Appointment Date 9/4/2025

Maintenance service ☒

Repairs ☒

Replacement Parts ☒

Service Amount \$10,000

* Vehicle number plate TN23TS2344 Please enter valid number

We hit a snag. Review the following fields: Vehicle number plate

Cancel Save

- **Billing & Feedback:** Ensures service rating is between 1 and 5.

Garage Manageme... Customer Details Appointments Service records Billing details and feedback Reports Dashboards * Home X

Billing details and feedback bill-001

Related Details * = Required Information

Billing details and feedback Name Name bill-001 Owner Dharani S

Service records ser-006 X

Payment Paid \$10,000

* Rating for service 6 rating should be from 1 to 5

Payment Status Completed

Created By Dharani S, 9/5/2025, 9:14 PM By S, 9/5/2025, 9:14 PM

We hit a snag. Review the following fields: Rating for service

Cancel Save

Duplicate Rules

Duplicate rules were configured on **Customer Details** to **prevent duplicate records** based on **email** and **phone number**.

Apex Trigger

A custom **Apex Trigger** was written on the **Appointment** object to:

- Distribute **service charges** based on selected services.
- Update **total billing** automatically.

Reports & Dashboards

- **Reports:** Service information reports grouped by:
 - Customer
 - Appointment Date
 - Service Status
 - Payment Status

REPORTS	Report Name	Description	Folder	Created By	Created On	Subscribed
Recent	New Service Information Report		Garage Management Folder	Dharani S	8/29/2025, 2:45 AM	

Report: Service information
New Service Information Report

Total Records: 2

	Customer Name	Appointment Name	Service records Name
1	dharani	app-008	ser-005
2	Vegashini	app-009	ser-006

- **Dashboards:** Visual dashboards were created to display:
 - Service ratings
 - Payment summaries

Dashboard
Customer review

⚠ Last refreshed 4 days ago. Refresh this dashboard to see the latest data.
As of Sep 1, 2025, 8:00 PM Viewing as Dharani S

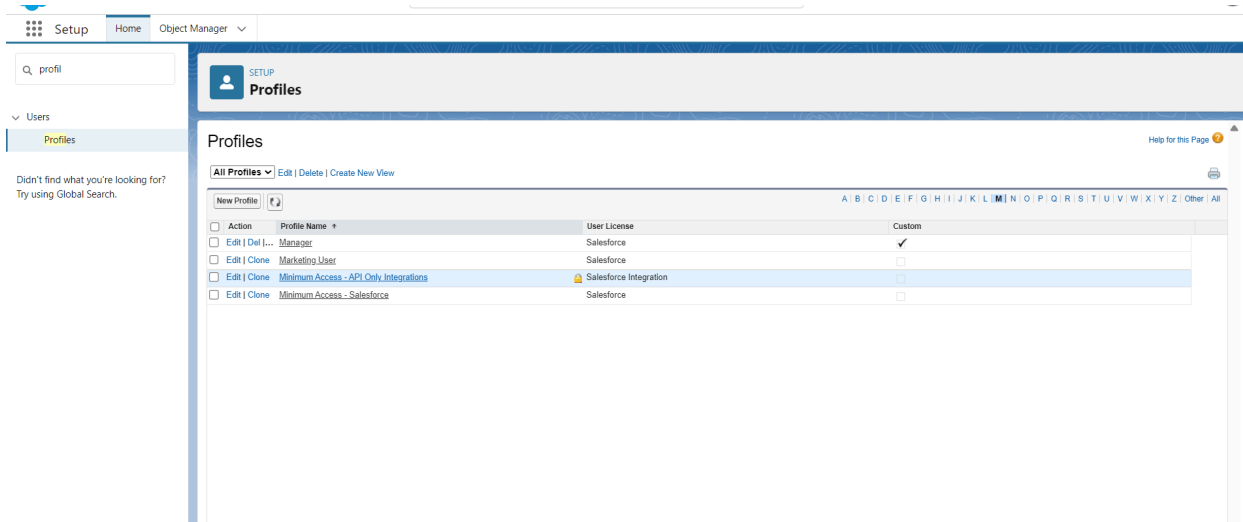
New Service information Report

Customer Name ↑	Appointment Name	Service records Name
dharani	app-008	ser-005

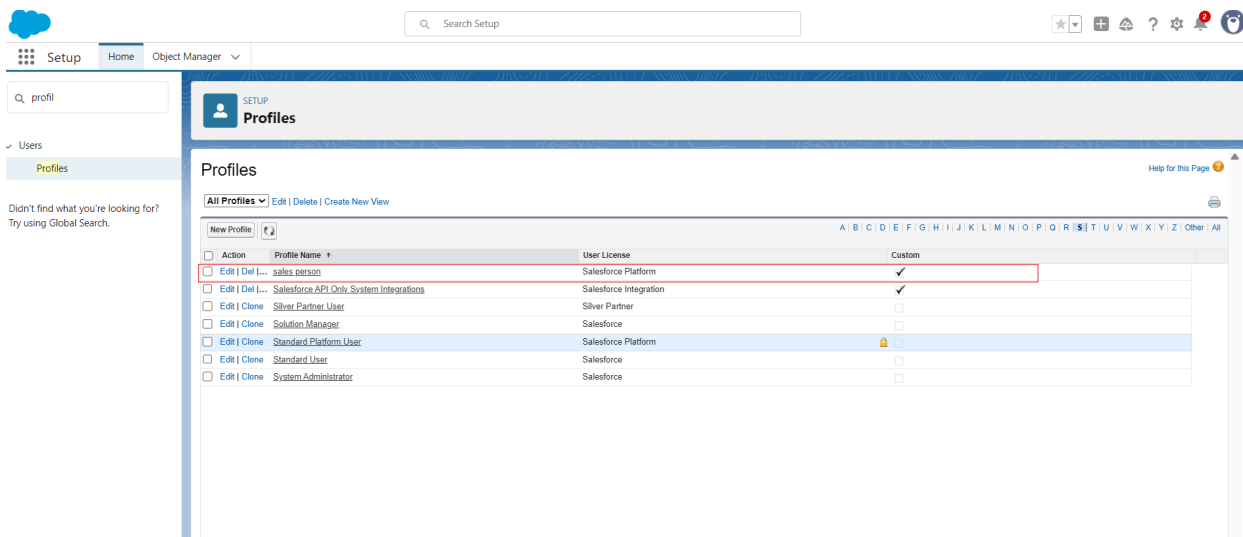
View Report (New Service Information Report) As of Sep 1, 2025, 8:00 PM

Profiles, Roles & Sharing Settings

- **Profiles:**
 - **Manager Profile:** Full access to all objects.



- **Salesperson Profile:** Limited access to service-related tasks.



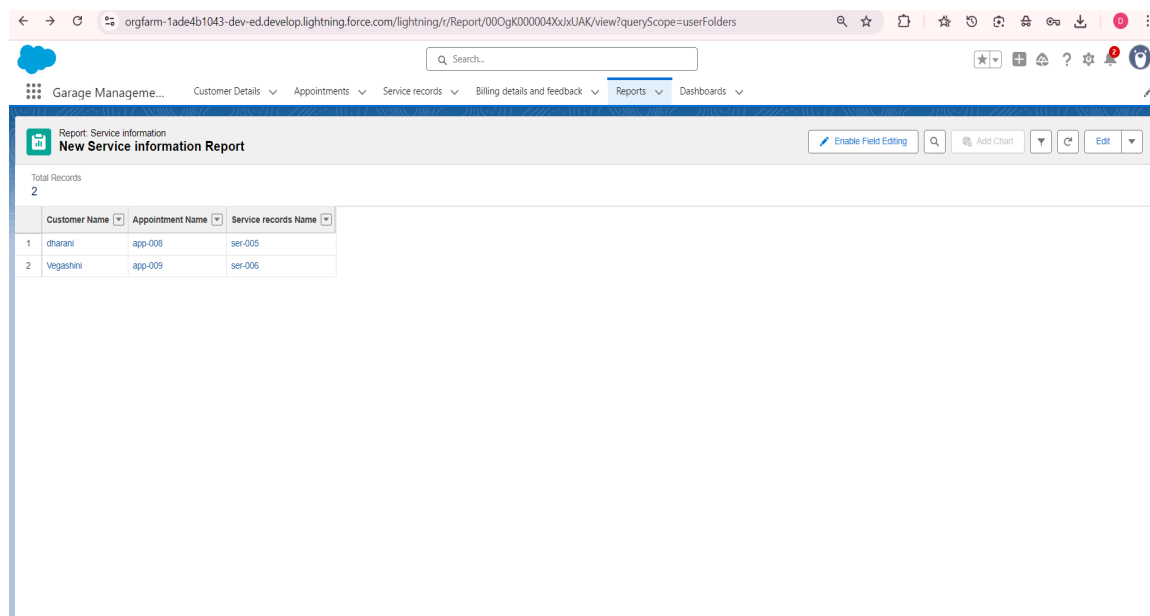
- **Role Hierarchy:** Managers at the top → Salespersons reporting under managers.
- **Organization-Wide Defaults (OWD):** Set to **private** for **Service Records**.
- **Sharing Rules:** Configured to **grant managers** access to team records.

4. Functional & Performance Testing

Testing was performed to ensure accurate functionality:

- Tested **validation rules** by entering incorrect or incomplete data.
- Verified **automation flows** by updating records and observing automatic updates.
- Checked **reports and dashboards** with sample records.
- Performed **role-based testing** by logging in as **Manager** and **Salesperson**.

5. Results



The screenshot displays a Salesforce Lightning interface. At the top, there's a navigation bar with tabs for 'Garage Manage...', 'Customer Details', 'Appointments', 'Service records', 'Billing details and feedback', 'Reports', and 'Dashboards'. The 'Reports' tab is active, showing a report titled 'New Service information Report'. Below the report title, there's a table with two columns: 'Customer Name' and 'Appointment Name'. The table contains two records: 1. Customer Name: dharani, Appointment Name: app-008, Service records Name: ser-005. 2. Customer Name: Vegashini, Appointment Name: app-009, Service records Name: ser-006.

	Customer Name	Appointment Name	Service records Name
1	dharani	app-008	ser-005
2	Vegashini	app-009	ser-006

- Tabs for **Customer Details, Appointments, Service Records, Billing & Feedback**
- Flow form for **appointment creation**
- Reports showing **service records and payments**
- Dashboards displaying **service ratings** and **payment analytics**

6. Advantages & Disadvantages

Advantages

- Streamlines garage operations
- Automates **appointment booking, billing, and service tracking**
- Enhances **customer satisfaction** through feedback integration
- Provides **real-time reports** and **dashboards** for better decision-making

Disadvantages

- Requires **internet connectivity**
- Needs **Salesforce knowledge** for initial setup
- Automation accuracy depends on **correct data entry**

7. Future Enhancements

- **Chatbot Integration** for **real-time customer support**
- **AI-based predictive maintenance** for suggesting future services
- **Mobile App Integration** for **on-the-go booking and tracking**

8. Conclusion

The **Garage Management System** built on **Salesforce CRM** successfully manages the **end-to-end garage operations**.

With the help of **custom objects, automation flows, validation rules, reports, and dashboards**, the system ensures:

- **Efficiency**
- **Accuracy**
- **Customer satisfaction**

Future enhancements like **chatbots**, **AI suggestions**, and **mobile apps** will further **improve the user experience** and **business growth**.

9. Appendix

Sample Apex Trigger

```
trigger ServiceChargeTrigger on Appointment__c (before
insert) {
    for(Appointment__c app : Trigger.new) {
        app.Total_Service_Charge__c =
app.calculateCharges();
    }
}
```

Sample Scheduled Class

```
global class DailyServiceReminder implements Schedulable {
    global void execute(SchedulableContext sc) {
        sendDailyReminders();
    }
    public static void sendDailyReminders() {
        List<Customer__c> customers = [SELECT Id, Email__c
FROM Customer__c];
        for(Customer__c c : customers) {
            Messaging.SingleEmailMessage mail = new
Messaging.SingleEmailMessage();
            mail.setToAddresses(new String[]{c.Email__c});
            mail.setSubject('Garage Service Reminder');
            mail.setPlainTextBody('Please check your
appointment and billing details.');
```

```
            Messaging.sendEmail(new
Messaging.SingleEmailMessage[]{mail});
        }
    }
}
```

